

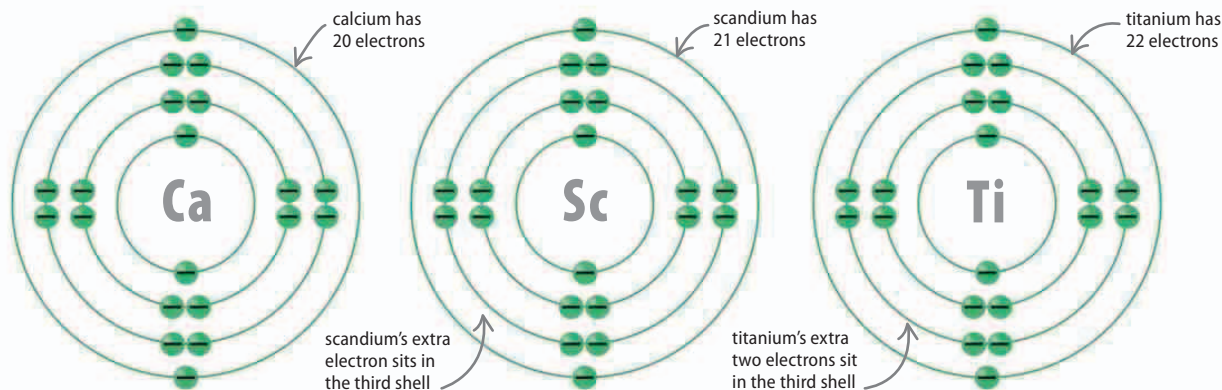
Transition metals

THE TRANSITION METALS ARE GROUPED IN A BLOCK THAT FORMS THE CENTER OF THE PERIODIC TABLE.

The transition metals make up a block from Groups 3 to 12 in the periodic table. They have distinct chemical properties because of the unique way that their electrons are arranged inside the atoms.

Inner and outer electrons

The transition elements only have one or two outer electrons. This is because they can put more than eight electrons in the shell below the outermost shell. So as the atomic number of the elements increases along each period in this block, the extra electrons are not held in the atoms' outer shells, but are put in the next shell down, which can hold up to 18 electrons. This is known as back-filling.



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- ◀ 112–113 Ionic bonding
- ◀ 118–119 Understanding the periodic table
- ◀ 120–121 Alkali metals and alkali earth metals
- Redox reactions **132–133** ▶

▼ Adding electrons

Calcium is not a transition element. It has two outer electrons and eight in the next shell down (the third). However, next along in the periodic table is scandium—the first transition element. It has one more electron than calcium, but it sits in the third shell, so a scandium atom still has two outer electrons. Similarly, titanium has two outer electrons but ten in the third shell.

Different charges

Like other metals, transition elements lose their outer electrons easily to form positive ions. However, transition metals can then continue to lose electrons from the next shell down and so can form ions with a number of different charges, or oxidation states. An ion's oxidation state indicates how many electrons have been lost or gained: every electron lost increases the oxidation state by one; +2 means two electrons have been lost (see page 132). For example, manganese (Mn) forms ions with five common charges.

Oxidation state	Electrons lost by manganese
+2	two outer electrons
+3	two outer electrons and one inner electron
+4	two outer electrons and two inner electrons
+6	two outer electrons and four inner electrons
+7	two outer electrons and five inner electrons

REAL WORLD

Most useful metals

Transition metals are less reactive than the alkali and alkali earth metals in Groups 1 and 2. Because of this, they have been used in technology for thousands of years. Iron is the most common transition metal. It is a very strong construction material. Nickel, another transition metal, is used in many coins.

